



Review

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ABSTRACT

Trademark opposition occurs when a newly filed trademark application is challenged for potential conflict with earlier rights, requiring a detailed comparison of texts describing protected goods or services. Using a dataset of 107,570 EUIPO records (2012–2024) describing office practices and decisions on pairs of goods and services, this paper presents different models to automatically predict the legal decision on similarity. The first contribution of our approach is to fine-tune embeddings from transformers and use them to vectorize the textual description of protected goods and services. The second contribution concerns a proposal of a hybrid neural network model that incorporates both the model embedding and categorical variables describing the context in which the legal decision was made. The goal is to provide a model able to learn both the semantics of the text description and the institutional rules for determining the similarity of the protected goods and services.

We show that our model significantly improves on the state of the art, with more than 90% of F1-score, where previous approaches do not exceed 70%. We also propose lighter more explainable models (Decision Trees, Logistic Regression, Random Forest), with F1-score between 80% and 90%, that can be used by experts to justify justice decisions.

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1. Introduction

1.1. Context

A trademark is an industrial property right that protects a sign (e.g., a name, logo, etc.) for products and services specified in a list of items (words or groups of words). For example, Hugging Face, Inc.'s registered trademark protects the "Hugging Face" word mark for use in relation to products and services, such as: "Downloadable computer software development tools and frameworks for developing software applications in the fields of data science, artificial intelligence, machine learning, modeling, and deep learning".

The purpose of a trademark opposition is to demonstrate that a newly applied for trademark infringes on prior one. In order to do that, where the marks are not identical, it is necessary to demonstrate how the two trademarks are similar and likely to cause confusion among consumers. This involves a comparison of the signs (brand name and logo), on the one hand, and a complex analysis of the overlap in terms of products and services, on the other. For example, it would be necessary to determine how the item "jewelry" is similar to the item "bracelets", or, when defending an opposition, to show that the item "ballpoint pens" is different from the item "balls for pens".

A trademark opposition may require the comparison of hundreds of products and services, based on evaluation criteria set by the applicable laws and regulations, and case law (past decisions of registration offices or courts). For example, the comparison of two trademarks each protecting 200 goods and services requires the evaluation of 40000 pairs of items. Practitioners specializing in trademark opposition need specific tools to quickly match and compare products and services.

Numerous studies are using artificial intelligence to study trademark similarity. Most of them concern comparisons of logos with a computer vision model [1–7], or the brand name from a syntactic and phonetic point of view, or a combination of both [8–10]. The automatic analysis of semantic information contained in the texts describing products and services is less present in the bibliography. Recently, a number of studies [11–13] investigated the potential of language foundation models in determining the distinctiveness of trademarks, i.e. the extent to which the brand name describes the products and services. In addition to mediocre results, this approach poses the problem of model ownership and the security of transmitted information. In 2024, [14] trained their own NLP deep learning models (Fine-tune of a DistilBert and DistilRoberta models) using a dataset with over 1.5 million trademark registrations to determine the distinctiveness of a brand. Since these models only take text as input, they concatenated all the information from their dataset (brand name, description of goods and services, Nice class, ...) into a paragraph to train their model.

To the best of our knowledge, only [15] have focused their study on the comparison of products and services to establish the similarity of trademarks. Their approach consists in vectorizing items and then using a cosine or Tanimoto distance similarity measure. The vectorizing models (embeddings) used are specific to word vectoring, and only minimally integrate sentence context. The comparison of items is based entirely on the calculated similarity measure, and does not incorporate any other information, such as the registration office or the origin of the decision. Their study focuses exclusively on Class 45 of the Nice classification. This classification was instituted by the Nice Agreement in 1957, and a new edition is currently published

every year (<https://www.wipo.int/classifications/nice/fr/>). This classification is an international classification of goods and services for the purposes of trademark registration. Preliminary versions of our work were presented in French conference proceedings [16]; [17], while this manuscript provides the full three-tier pipeline, expanded experiments, and governance-oriented workflow discussion.

1.2. Research objectives

The objective of this study is to build upon the work initiated by [15] in the field of automatic similarity detection. The approach adopted will be close to that employed by [14] in the context of distinctiveness. The main contributions of our work are:

An efficient embedding model. The first objective is to change the vectorization of items used by [15]. We replace the word embedding model with sentence embedding models, fine-tuned from transformers models with the dataset. The cosine similarity measure is then compared to a threshold for the item similarity prediction. We suggest a formula for automatically selecting an optimal threshold. We compare different models, in particular those used by [14]. This first approach significantly outperforms the state of the art in terms of prediction performance. A comparison of model performance in relation to their size shows that very large embeddings (e.g., Stella, 8192 dimensions) generate only about a 1% gain, raising questions of cost-effectiveness that are addressed in the third objective.

Integration of contextual information. The second objective is to determine a model able to predict court decisions, regardless of the Nice class or the office making the decision. We propose a new model that incorporates not only embeddings, but also categorical features describing the context in which the legal decision was made. In trademark opposition, the question is not only whether two item descriptions are linguistically close; it is whether the legal decision on similarity should be positive or negative under institutional practice. This legal decision is determinant for the validity of a contested mark. Our formulation therefore separates a semantic substrate (the meaning of the goods/services as captured by sentence embeddings) from institutional context (office, origin, and class signals) that calibrates how that semantic signal is adjudicated. The model is trained on pooled decisions across offices to learn a common representation of semantic relatedness, while contextual features enable it to capture, for each office, the legally grounded mappings by which similar factual situations are converted into decisions.

This approach takes into account more information from the dataset than a simple cosine measure or the Tanimoto distance used in the previous work of [15]. An analysis of feature importance reveals that contextual variables significantly impact prediction. Contrary to [14], we did not incorporate these additional variables within a global embedding to avoid diluting their impact in the textual description of the items. The large language model is only used to vectorize texts describing products and services. The contextual features, being categorical, are encoded with one-hot-encoding. In addition, we have built features to take into account interactions between Nice classes. Text embedding, encoded contextual variables and interactions are then used as inputs for a neural network that we trained on our dataset.

Smaller and explainable model. The obtained model has good performances but is rather big and non-explainable. For this reason, as a third objective, we also propose a slightly less efficient alternative, which produces lighter and partially explainable models, a critical aspect for jurists, with a tidy decreasing of performances.

Table 1
Extract of the dataset.

PID	Nice Classes : Items	Office	Origin	Decision
00..39	11 : Apparatus for ventilating 19 : Rigid pipes, not of metal, for building	EM (EUIPO)	OP	Similar
		Italy (UIBM)	OP	Dissimilar
		Malta (CD IPRD)	OP	Dissimilar
00..34	7 : Aeroplane engines 12 : Vehicles	EM (EUIPO)	OP	Similar
00..20	11 : Acetylene generators 11 : Refrigerating apparatus	EM (EUIPO)	OP	Dissimilar
00..56	12 : Bicycles 12 : Motorcycles	Sweden (PRV)	NC	Low similar
00..06	3 : Tooth cleaning preparations 21 : Floss for dental purposes	Sweden (PRV)	CJ	High similar

Section 2 defines the dataset and the validation protocol used across all experiments. Section 3 evaluates embedding-based cosine rules and threshold selection. Section 4 introduces the hybrid model that combines embeddings with contextual variables and reports the corresponding gains. Section 5 derives reduced, more explainable models and quantifies their trade-off against the hybrid model. Section 6 discusses workflow integration and governance constraints. Section 7 concludes and outlines future work.

2. Dataset and process overview

The European trademark office, called EUIPO, is making available data concerning practices or decisions on pairs of items taken by various European Offices (<https://euiipo.europa.eu>) over the period 2012 to 2024. The dataset used in this paper includes 107,570 practices and decisions entered by: The European Union Intellectual Property Office (EUIPO) publishes data concerning practices and decisions on pairs of items from EUIPO and national offices (<https://euiipo.europa.eu>) over the period 2012 to 2024. The dataset used in this paper includes 107,570 records entered by:

- a pair of items describing the products or services (number of words : $min = 1, max = 23, mean = 3.5$),
- the Nice classification for each of the two items (45 values),
- the court or trademark office that rendered the verdict (16 values),
- the origin of the decision, e.g. Court of Justice, Board of Appeal, Office Practice, ... (6 values),
- a legal decision of similarity between the two items given by a court or trademark registration office.

These records describe office and court treatment of item pairs in opposition and examination practice; they are therefore not limited to trademarks that completed registration.

The legal decision is composed of several levels (Identical, Similar, High Similar, Low Similar, Dissimilar). The aim is to produce an automatic alert on potential similarity, so we grouped all levels of similarity into a single class. The target variable is then binary, denoting whether the items are similar (positive class) or dissimilar (negative class).

As illustrated in the dataset extract (Table 1), a pair of items can appear multiple times in the dataset depending on the office and the origin (PID 00..39). It is important to note that the justice decision may not be consistent. For instance, the Maltese and Italian offices have concluded that ‘Apparatus for ventilating’ and ‘Rigid pipes, not of metal, for building’ are not similar, a decision that is at odds with

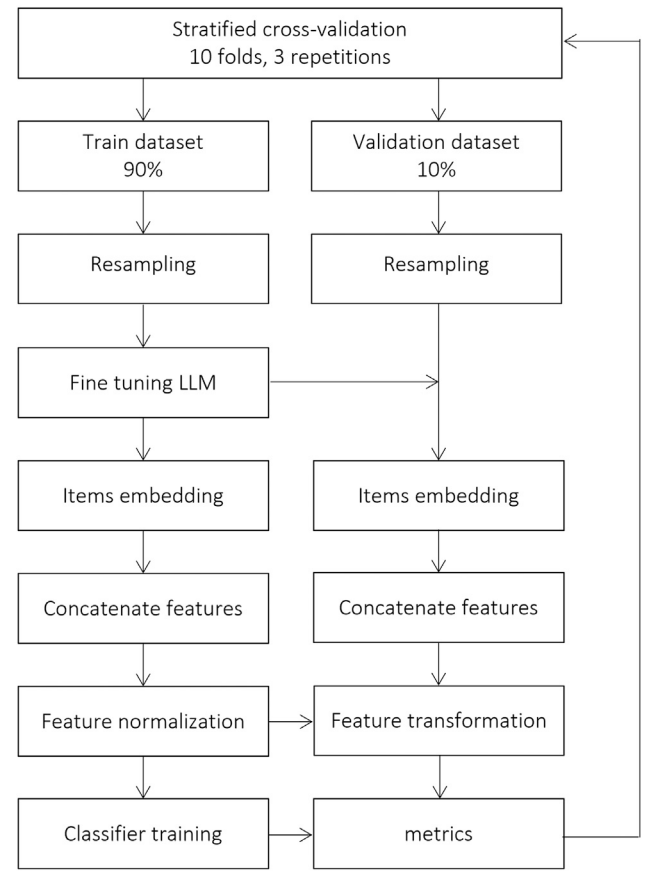


Fig. 1. Stratified cross-validation process.

that of the European office. The detection of similar items is not only based on the semantic similarity of the items, but can also be influenced by additional parameters such as the office that made the decision or the Nice classification categories of the items. The aim is to build a model able to predict the justice decision, either solely on the basis of the semantic information contained in the items or by introducing additional information on the context in which the decision was made.

In the following sections, all models are compared under the same validation protocol. The target classes are imbalanced (Similar = 15,846, Dissimilar = 91,724), so each run includes random down-sampling of the dissimilar class to build balanced training and validation subsets (31,692 records in total). The process has three sources of randomness: class balancing, train/validation split, and model training. To quantify robustness, we run a stratified cross-validation with 10 folds and 3 repetitions (30 runs), as shown in Fig. 1. For each run, embeddings are fine-tuned only on the corresponding training subset, classifiers are trained with the same split, and metrics are computed on held-out validation data. Reported values are means over the 30 runs, and significance tests are applied to run-level metrics.

In order to compare the models, taking into account the uncertainty due to the different sources of randomness, we will use the statistical tests :

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_d \text{ vs } H_1 : \exists i \neq j, \mu_i \neq \mu_j$$

where μ_i represents the mean F1 score (or other metric) (Student's test if $d = 2$ and ANOVA test if $d > 2$). If the p -value of the test is greater than 5%, then we can conclude that there is no significant difference in performance between the models. We repeat the cross-validation 3 times in order to obtain a sufficiently large sample (30 observations of the usual metrics) to apply statistical tests.

3. The usual approach with the cosine similarity measure

[15] used the item embeddings to assess the similarity. They recommend using the sentence vectorization model Doc2Vec (d2v) trained on their dataset, or the word vectorization model Word2Vec (w2v) trained on Google news, with the cosine similarity measure. In this work, we suggest to replace Doc2Vec or Word2Vec by an embedding model based on transformers [18]. We compare five large language models (LLM3, LLM12, mpnet, DistilRoberta and Stella) given in Table 2 that have been fine-tuned on the data described previously. We select different size representing state-of-the-art of LLM from original sentence transformers or more recent embedding on massive data [19]. The initial LLMs have been learned with different datasets in English, described in previous citations [18,19]. With all these initial LLMs, we proceed to the same fine-tuning, consisting in a selection of data with the cross-validation process described in Fig. 1, and following hyperparameters : batch size= 16, loss function = cosine similarity, warmup = 0.01 and epochs = 1 or = 10. In the appendix we provide the results of a sensitivity analysis to determine the influence of the hyperparameters on the F1 score. The number of epochs emerges as the most influential factor. The F1 score on the validation set obviously increases with the number of epochs, reaching a stable level at 10 epochs. However, it should be noted that fine-tuning a LLM with 10 epochs is computationally expensive. For this reason, the two alternatives of 1 or 10 epochs are compared in the rest of the paper.

The usual approach to compare two items uses the cosine similarity measure with the simple rule:

$$\text{If } \text{Cosine}(E_1, E_2) > \text{threshold then Item1 and Item2 are similar} \quad (1)$$

where E_1 and E_2 are the embeddings of Item 1 and Item 2.

The only hyperparameter of this model is the threshold. Fig. 2 shows model metrics as a function of the threshold for the LLM3 model. As the threshold increases, recall decreases and specificity increases. Recall is the true positive rate and specificity is the true negative rate, i.e. the percentage of similar and dissimilar pairs that are well classified, respectively. Both metrics are important for trademark opposition. In Fig. 2, a low threshold favors recall but increases false positives, which creates unnecessary review workload. By contrast, a high threshold favors specificity but increases false negatives, which may hide truly similar pairs and increase downstream legal risk.

$$\frac{|\text{Recall} - \text{Specificity}|}{\text{Recall} * \text{Specificity}} \quad (2)$$

The absolute value between recall and specificity is used to find the best compromise, while the denominator is used to discriminate threshold values for which both metrics are very small. In the appendix, we explain why we chose this index rather than the traditional Youden's J index.

Fig. 3 represents the evolution of (2) as a function of the threshold. For LLMs and Doc2Vec models, the optimum threshold is around 0.5. Fig. 4 shows that a threshold near 0.5 effectively separates similar and dissimilar classes for LLM models. However, the distribution of cosine values for Doc2Vec indicates that no threshold cleanly separates the two classes. The optimal threshold value for Word2Vec is lower, around 0.3, but Fig. 4 still shows substantial overlap between classes. This initial observation is confirmed by the poor performance of these two models, as discussed below.

The performance comparison was carried out using the stratified cross-validation method described in Fig. 1. The usual metrics are calculated on the validation dataset. Fig. 5 gives the ROC curves and Table 2 gives the average F1 scores obtained with the optimal threshold. With balanced classes, a model that systematically classifies all items as similar achieves an F1 score of $2/3$ since $TP = FP = n/2$ and $FN = TN = 0$, then $\text{Recall} = 1$ and $\text{Precision} = 1/2$, where n is the number of classified examples. The F1 score of Doc2Vec is below this value and the best F1 score of Word2Vec is 0.69. In Fig. 5, Doc2Vec

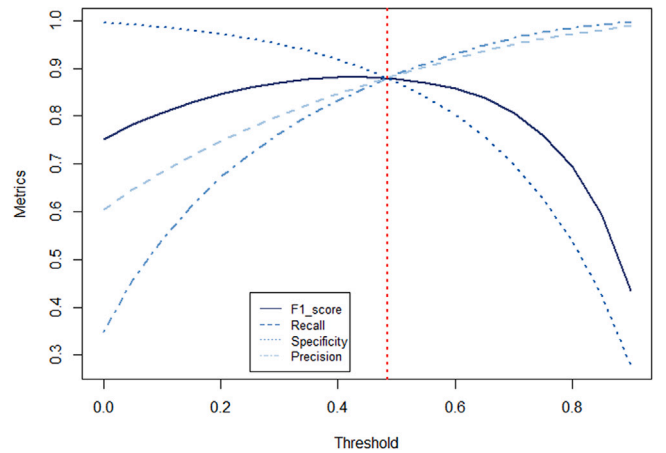


Fig. 2. Metrics as a function of the threshold for the cosine-rule with LLM3 embedding model trained with 10 epochs.

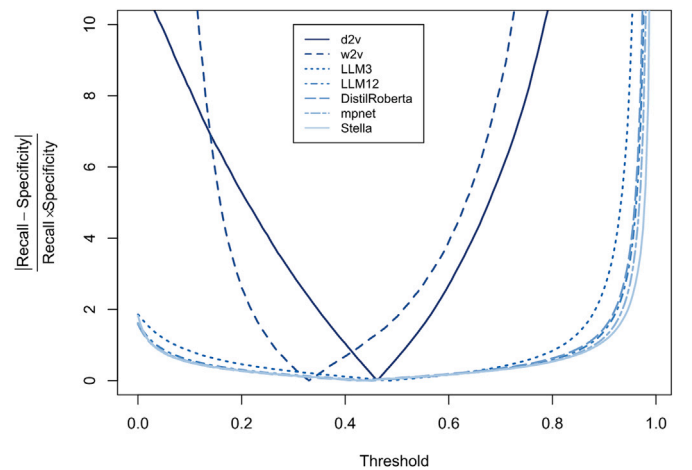


Fig. 3. Formula (2) as a function of the threshold for the cosine-rule with different embedding models (trained with 10 epochs for LLMs).

Table 2

Embedding models comparison for the cosine-rule (average of the 30 F1 scores)

Model	Embedding size	Model size	F1 score (1 epoch)	F1 score (10 epochs)
d2v	50	22 MB	0.438	/
w2v	300	100 MB	0.690	/
LLM3	384	61 MB	0.815	0.879
LLM12	384	120 MB	0.850	0.888
DistilRoberta	768	82 MB	0.842	0.885
mpnet	768	420 MB	0.861	0.890
Stella	8192	1.62 GB	0.870	0.891

is below the diagonal and Word2Vec is just above. By contrast, the F1 scores of the five LLM models are all much higher than $2/3$ and their ROC curves are well above the diagonal. So, using an LLM is a clear improvement on the approach proposed by [15]. The fact that the ROC curve of Doc2Vec is below the diagonal suggests that the labels of the target variable are reversed, and this observation is confirmed by the boxplots of cosine values in Fig. 4. Even if we reverse the rule in Eq. (1) (which is not mathematically correct), the ROC curve moves just above the diagonal and the best F1 score is 0.56 ($< 2/3$). The poor performance of Doc2Vec and Word2Vec is therefore explained by the overlap in cosine distributions.

Concerning the choice between the five LLMs, the ANOVA test shows no significant difference between their F1 scores for 10 epochs

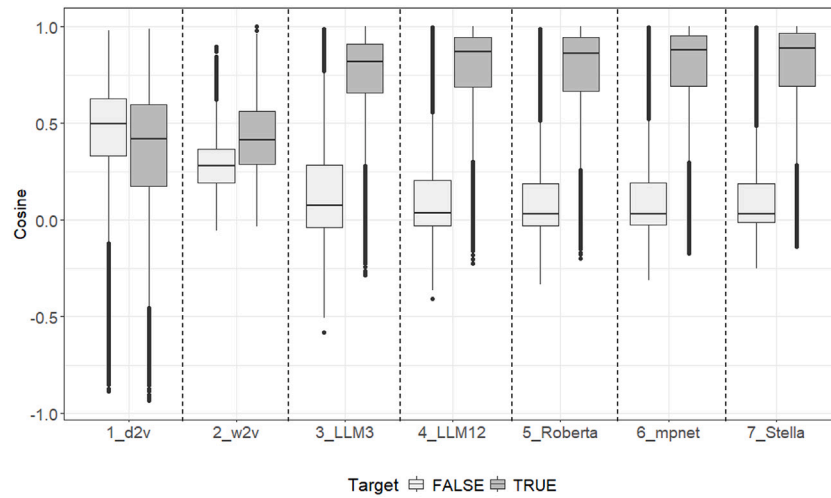


Fig. 4. Boxplots of the cosine values (LLMs trained with 10 epochs).

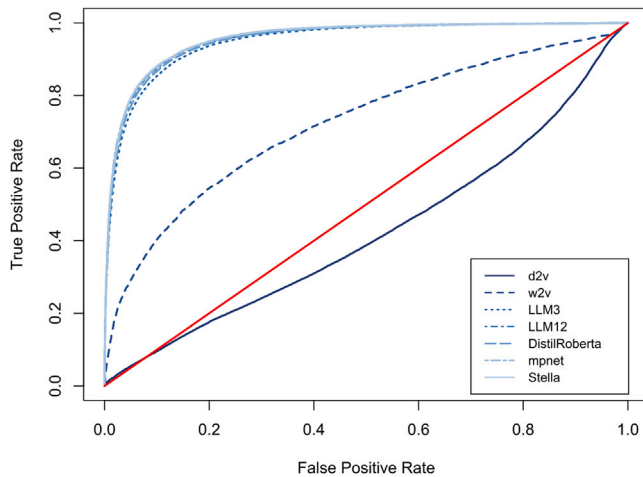


Fig. 5. ROC Curves to compare different embedding models (trained with 10 epochs for LLMs) for the cosine-rule.

(p -value $< 2.2e10^{-16}$) and a little difference with only 1 epoch (p -value = 0.1). However, even if the difference is not statistically significant, the F1 score increases with the embedding and model size (Fig. 6 and Table 2) before reaching a limit. Using a very large language model like Stella (1.2 GB) with an embedding size of 8192 only increases the F1 score by 1% for one epoch, and less than 1% for 10 epochs, compared with the mpnet model (420 MB) with an embedding size of 768. Conversely, we can see a difference in the F1 scores depending on the number of epochs. This discrepancy is particularly evident in models of smaller dimensions.

For the next section, we retain two model sizes: a small one LLM3 and a medium-sized mpnet, and the two alternatives 1 or 10 epochs for the fine-tuning. The goal is to see whether a small LLM with less computing time can achieve the same performance as a larger one or not, by incorporating information on institutional rules and practices into the model.

One limitation of this model is that classification is based solely on cosine value. This means that the model cannot distinguish contradictory decisions, such as PID 00...39 in Table 1. The LLM3 model, trained with 10 epochs, calculates a cosine value of 0.62 between the terms ‘Apparatus for ventilating’ and ‘Rigid pipes, not of metal, for building’. With a decision threshold around 0.5 (Fig. 3), the model predicts the similar class regardless of office. Consequently, this model

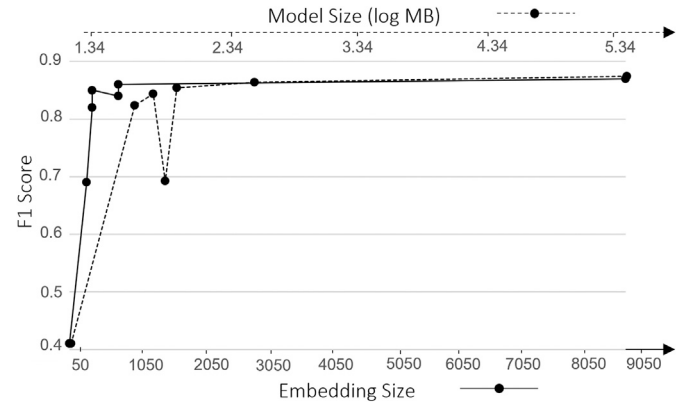


Fig. 6. F1 Score as a function of the model size and the embedding size (10 epochs).

is only correct for the European office in that example. This motivates the next section, where contextual institutional variables are added to resolve such contradictions.

4. A hybrid model with additional contextual information

The aim of this section is to improve the performance of the previous model by incorporating additional information from the dataset describing the context in which the legal decision was made. Specifically, we examine whether this information enables a smaller model (LLM3 fine-tuned with 1 epoch) to achieve performance close to a larger one (mpnet fine-tuned with 10 epochs). In addition, we seek to build a model capable of predicting different decisions based on context. The contextual information is stored in the features office, origin, Nice class of Item 1 (C1), and Nice class of Item 2 (C2).

4.1. Information in the contextual variables

To determine whether contextual variables provide information about the decision, we adjusted a model using only these features, without item embeddings. This model is a neural network with a single hidden layer of size 64 (called No Emb). The averages of the usual metrics are given in Table 3, and Fig. 9 gives the ROC curve. The average F1 score is 0.81, which shows that these features carry predictive information. This baseline quantifies the institutional signal and motivates combining it with item embeddings in the hybrid model below.

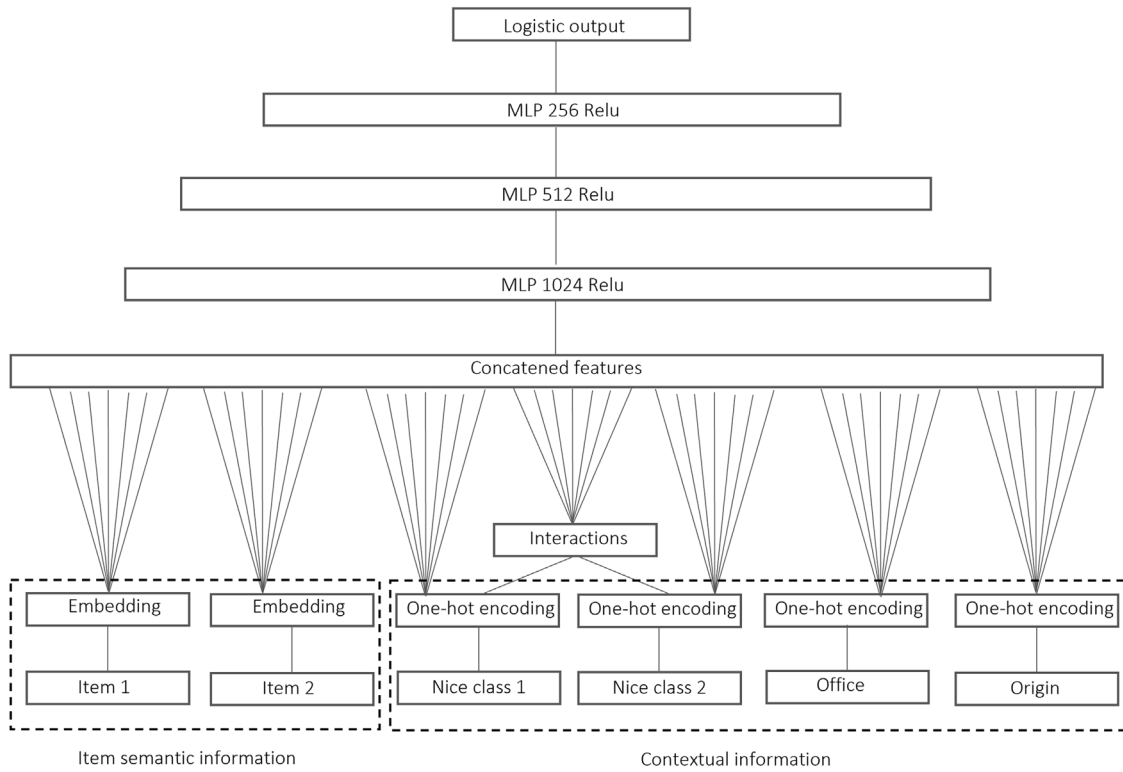


Fig. 7. Model diagram.

4.2. The hybrid model

The institutional features are categorical with 6 (origin), 16 (office), 45 (C1) and 45 (C2) and each level is binary encoded to obtain a total of 114 features.

Moreover, two items may be in different Nice classes and nevertheless be judged to be similar. For example, the European registration office considered the items “Aeroplane engines”, belonging to Class 7, and “Vehicles”, belonging to Class 12, as similar (PID 00..34 in Table 1). They argued that: “Vehicles in Class 12 cover aeroplanes. Engines in Class 7 cover engines for aeroplanes. These goods are complementary. The public may expect these engines to be produced by, or under the control of, the original manufacturer of the aeroplanes.” Conversely, two items in the same class may be dissimilar. For example, in Class 11, “Acetylene generators” and “Refrigerating apparatus” are considered as dissimilar by the EUIPO (PID 00..20 in Table 1). To take this information into account, we introduce class interaction into the model. If we denote c_i and c_j the encoded variables of Nice classes i and j , then interaction is defined as the variable $c_i + c_j$, i.e. the simultaneous occurrence of both classes. The number of interactions is 1035.

And finally, instead of summarizing the semantic information contained in the items by a simple cosine, we decided to consider the entire embeddings as features.

The schematic diagram of the model is shown in Fig. 7.

The components of item vectorization, encoded categorical variables, and class interactions are concatenated. The number of features in the new dataset depends on embedding size. The classification model is a neural network with 3 hidden layers of size 1024, 512, 256 and ReLU activation (Hybrid MLP). The number of classifier parameters depends on embedding size: 0.9M for LLM3 and 1.1M for mpnet.

4.3. The results

We compare this model with the usual approach based on cosine similarity, and with the model without embeddings. For traceability,

Table 3

Mean values of the usual metrics for LLM3 and mpnet embedding models trained with 1 or 10 epochs and different classifiers.

Embedding	Classifier	F1 Score	Recall	Spec	Prec
No Emb	MLP	0.807	0.807	0.809	0.809
LLM3 - 1 epoch	cosine-rule	0.815	0.81	0.82	0.82
	Hybrid MLP	0.897	0.899	0.896	0.897
	DT	0.852	0.854	0.850	0.850
	LR	0.845	0.841	0.849	0.848
	RF	0.845	0.844	0.848	0.847
LLM3 - 10 epochs	cosine-rule	0.879	0.882	0.876	0.877
	Hybrid MLP	0.902	0.903	0.900	0.901
	DT	0.892	0.892	0.895	0.892
	LR	0.901	0.901	0.893	0.901
	RF	0.897	0.897	0.886	0.897
MPNET - 1 epoch	cosine-rule	0.861	0.86	0.83	0.86
	Hybrid MLP	0.900	0.901	0.900	0.901
	DT	0.876	0.876	0.887	0.876
	LR	0.885	0.885	0.886	0.885
	RF	0.879	0.879	0.878	0.879
MPNET - 10 epoch	cosine-rule	0.890	0.895	0.883	0.885
	Hybrid MLP	0.906	0.907	0.905	0.906
	DT	0.902	0.902	0.901	0.902
	LR	0.899	0.899	0.904	0.899
	RF	0.899	0.899	0.898	0.899

each line of Table 3 corresponds to one model family evaluated with the same 30-run protocol defined in Section 2, and Figs. 8 and 9 visualize the same run-level outputs (F1 distribution and ROC behavior).

- Does introducing contextual variables improve classifier performance?

For LLM3 and mpnet models, the p-values of the Student’s test that compare the F1 scores obtained previously with the item embeddings only (Table 2) and those obtained with the hybrid MLP classifier (Table 3) are very small (less than 1%). The incorporation of contextual

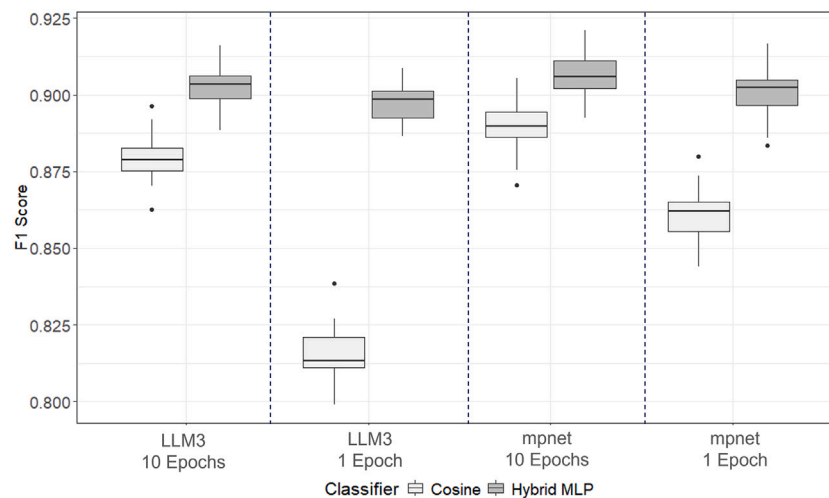


Fig. 8. Box-plots of F1 scores for the cosine-rule and hybrid MLP classifiers, LLM3 and mpnet embedding models and for the model without embedding.

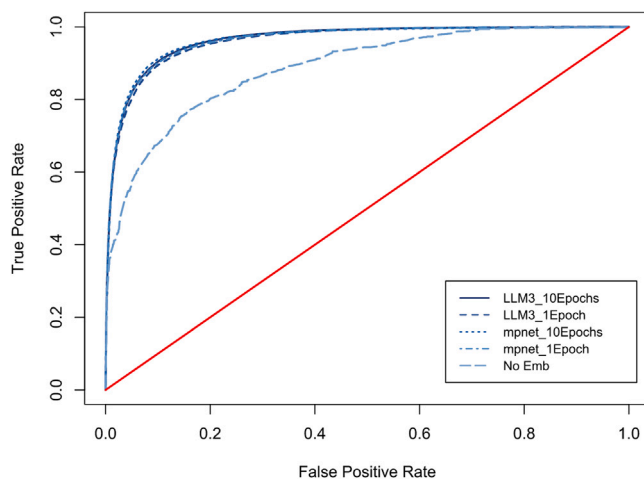


Fig. 9. ROC Curves to compare LLM3 and mpnet embedding models trained with 1 or 10 epochs in the hybrid model.

variables has clearly a significant impact on enhancing performance, irrespective of the embedding model employed. For a fine-tuning with 1 epoch, the various metrics have increased substantially, from 82% to 90% for LLM3, and from 86% to 90% for mpnet. For a fine-tuning with 10 epochs, the increase in metrics is less pronounced even if it is statistically significant. This conclusion is reinforced by Fig. 8. The box-plots of the F1 scores obtained with the cosine-rule are systematically below the box-plots obtained with the hybrid MLP classifier.

- Does incorporating contextual variables enable smaller models to achieve the same performance as larger ones?

For the hybrid MLP classifier, the p -value of the ANOVA test that compares LLM3 and mpnet embedding models fine-tuned with 1 or 10 epochs is 6×10^{-7} (equivalent to $6 \times 10^{-5}\%$). This indicates that there is a statistically significant difference in the F1 scores of the four models. However, the difference between the most cost-effective model (hybrid MLP with LLM3 fine-tuned with 1 epoch: F1 score=0.897) and the most expensive model (hybrid MLP with mpnet fine-tuned with 10 epochs: F1 score=0.906) is less than 1%. This conclusion is confirmed by the ROC curves, which are almost identical in Fig. 9. Consequently, there is no benefit to be gained from the use of a large embedding model fine-tuned with 10 epochs in order to increase predictive scores with the

hybrid MLP classifier proposed in this work. Incorporating institutional rules with the contextual variables reduces the size of the model and the computational time of the fine-tuning step.

4.4. Importance of the office

The question now is which of the variables we have introduced into the model have a real impact on classification? SHAP values [20] is a common way to understand how model features impact predictions. The SHAP values have been computed for each example of the validation dataset and the 15 most important values are represented in Fig. 10. Among the most important variables are offices and embeddings. The first values of origin, interactions and Nice class only come in 37th, 673rd and 800th positions respectively. We can conclude that it is the addition of the office variable that increases the performance of the LLM model. To confirm this conclusion, we adjusted the hybrid MLP model without the office feature. We observed a significant decrease in the F1 score of 2.6%. On the other hand, the inclusion in the model of the Nice classes and the origin of the decision seem to have very little impact on the prediction.

The fact that the office variable has a high importance could indicate a bias according to who made the judgment. Fig. 11 represents the F1 score per office. We can see that the model performs well regardless the office, except for Malta. This exception cannot be explained by Malta's under-representation in the dataset. Indeed, the model's performance is not linked to the number of examples per office. For example, the Finnish office obtains an F1 score equal to that of the European office, even though it represents only 1% of the dataset, i.e., less than Malta. However, we noted that there were 22% of divergent decisions between the Maltese office and the European office, compared to, for example, 3% of divergent decisions between the Spanish office and the European office. As the European office is regarded as a reference for court decisions, the Maltese exception may be explained by its divergence from the European office.

If we consider the example of PID 00...39 decisions from Table 1, the LLM3 model, trained over 10 epochs, calculated a cosine value of 0.62. This resulted in a prediction in the similar class, regardless of the office. The Hybrid MLP model now predicts a probability (averaged over 10 repetitions) of being similar of 0.96 (standard deviation = 7×10^{-2}) for the European office, 0.77 (standard deviation = 3×10^{-1}) for the Italian office, and 0.06 (standard deviation = 6×10^{-2}) for the Maltese office. The model is now capable of predicting inconsistent decisions.

We therefore built a model that learns both the semantics of items and the institutional rules for determining the similarity of products and services. This approach outperforms the one proposed by [15] in

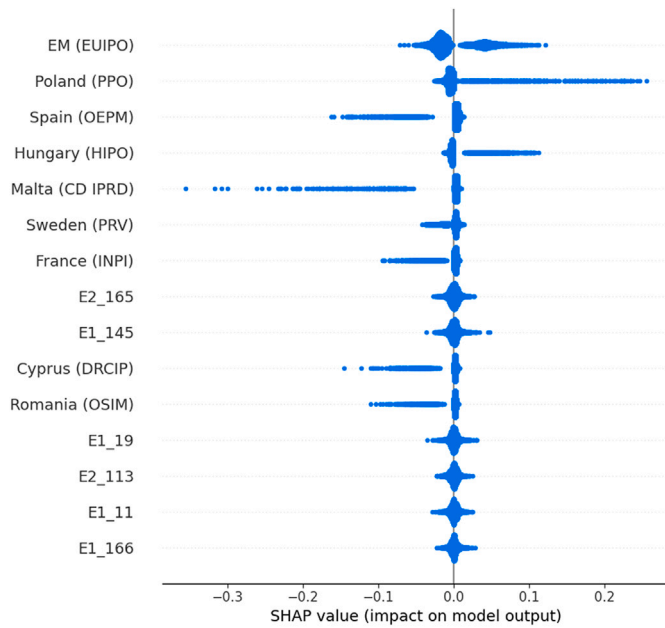


Fig. 10. Example of feature importance in the hybrid model.

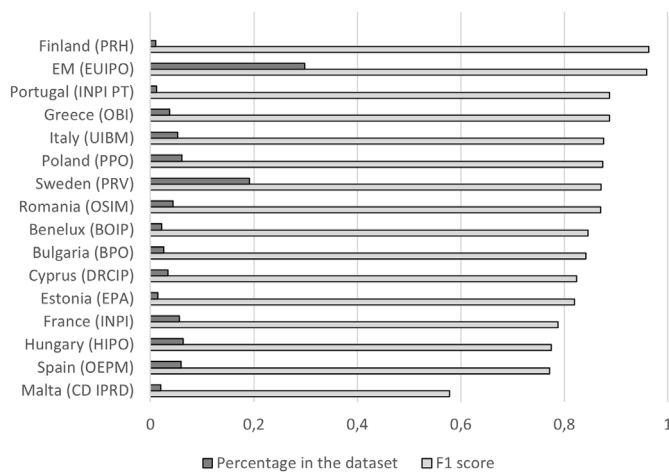


Fig. 11. F_1 score per office for the hybrid MLP classifier with the LLM3 embedding model fine-tuned with 10 epochs.

terms of performance, but also because it can be applied regardless of the Nice class, the office, or the origin. Our approach differs from the one developed by [14] in the context of distinctiveness. We did not include all the information in a single text (which is then processed by an LLM) in order to better highlight contextual variables, particularly the office. Including the name of the office in a text, that could be long with the names of the items, might have diminished its decisive role in prediction.

5. A compromise between additional information and model size

The previous paragraph showed that integrating context variables significantly improves performance. However, the resulting model is heavy and non-explainable. Explainability is the utmost importance when classification leads to a legal decision. In this section, we consider smaller models that are explicable or partially explicable. To do this, we reduce the dimension of the inputs. Item embeddings are replaced by their cosine similarity score, and interactions between Nice classes by a binary variable Diff: 1 if the classes are similar and 0 otherwise. The

two original item embeddings are high-dimensional vectors (hundreds of dimensions), which are difficult to interpret in legal contexts. By replacing them with a single cosine value the dimensionality of the model is dramatically reduced, while at the same time increasing interpretability. The cosine similarity is intuitive (ranging from 0 to 1) and can be easily explained to non-technical stakeholders, such as courts or trademark offices. The model shown in Fig. 12 now has only 6 inputs.

For the classifier model, we compared

- a logistic regression (LR),
- a decision tree (DT) with the Gini criterion and a maximum depth of 10,
- a random forest (RF) with 100 trees and 2 variables tested at each node.

In Fig. 13 and Table 3 (the ROC curves are given in the appendix), we can see that, for each embedding model, the three reduced models DT, LR and RF give equivalent results (difference is less than 1%). Neither one outperforms the other, and their slight difference in performance depends on the embedding models. In all cases, the reduced models outperform the model using only the cosine rule. So, once again, we can conclude that there is value in introducing contextual variables to model institutional practices. If we now compare the best reduced model with the big hybrid model, we can see a significant difference for embeddings fine-tuned with a single epoch (4.5% for LLM3 and 1.5% for mpnet). On the contrary, for embeddings fine-tuned with 10 epochs, this difference is drastically reduced (0.1% for LLM3 and 0.4% for mpnet). This means that a reduced model can be used instead of a large model, on condition that the semantic information contained in the text of the items has been correctly learned by the embedding model. On the other hand, to learn the semantic information of the items correctly, it is not necessary to use a large embedding model. Indeed, for 10 epochs, the difference between the best reduced models between LLM3 and mpnet is 0.1%. The good performance of reduced models confirms the results presented by [21] on the classification of legal texts using random forests.

We can therefore conclude that using a reduced model with an LLM3 embedding model fine-tuned with 10 epochs is a good alternative to the large hybrid model. The advantage is that this model is explainable or partially explainable. Fig. 14 shows the importance of the variables obtained with an RF classifier. The results confirm those obtained with SHAP values, i.e. the most important variables are the cosine between the embeddings of the two items and the office that made the legal decision. For the RL classifier (Fig. 15), the information is more complete than that provided by the random forest. As with SHAP values, we can see exactly which modalities are important in the model. We find the same results as in Fig. 10: the most influential offices are EUIPO, PPO, HIPO, etc. However, unlike the SHAP values or the importance of RF variables, here the fact that the two items are not in the same class (level: DiffTRUE) is decisive in the model. Finally, the classifier that provides the most information is the decision tree, which explicitly gives the decision rules (Fig. 16). We can see that the order in which the variables appear reflects the importance of the variables established by the previous methods. If we consider the example of PID 00...39 decisions from Table 1, since the cosine value is 0.62, the three decisions are in the branch in the middle. The prediction then differs depending on the office.

6. Workflow integration

The models developed in this paper are intended to assist, rather than replace, legal reasoning in likelihood-of-confusion analysis. They provide quantitative similarity scores and office-specific predictions that can be embedded in existing procedures for clearance, opposition, and examination.

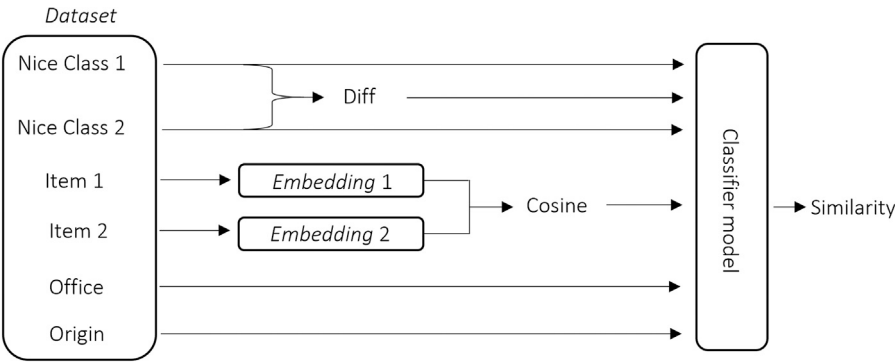


Fig. 12. Reduced model.

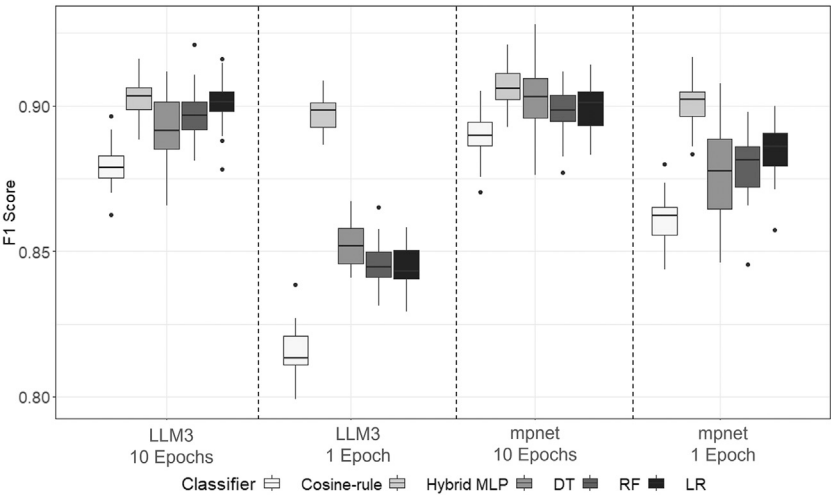


Fig. 13. Box-plots of F1 scores for DT, RF and LR classifiers and LLM3 and mpnet embedding models.

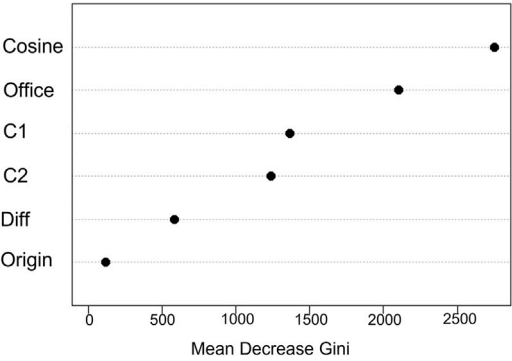


Fig. 14. Example of variable importance for the RF classifier with LLM3 embedding model fine-tuned with 10 epochs.

6.1. Users and decision points

Trademark attorneys are natural first users of these models. During clearance and pre-filing searches, candidate marks can be screened against earlier rights by scoring and ranking the pairs of protected goods and services. The ranking directs attention to those pairs that warrant detailed legal analysis. In opposition or invalidity proceedings, where two lists of several hundred items may yield tens of thousands of combinations, the same mechanism reduces the search space and helps concentrate written submissions on the combinations that carry

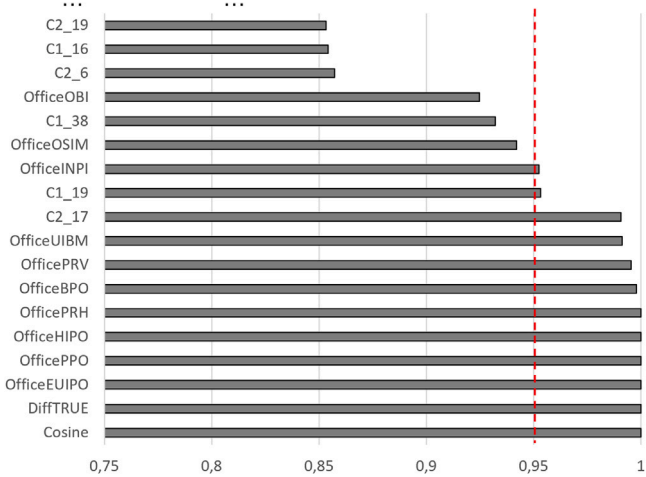


Fig. 15. Example of variable importance for the LR classifier with LLM3 embedding model fine-tuned with 10 epochs (1-p-value of the Wald's test).

the highest inferred risk of confusion. Traditionally, goods and services comparisons relied on coarse pre-filters based on Nice classes and labor-intensive manual inspection of the remaining pairs. The proposed models replace this coarse screening by a continuous similarity signal learned from past decisions, which allows faster and more precise prioritization of item pairs while remaining compatible with existing attorney workflows.

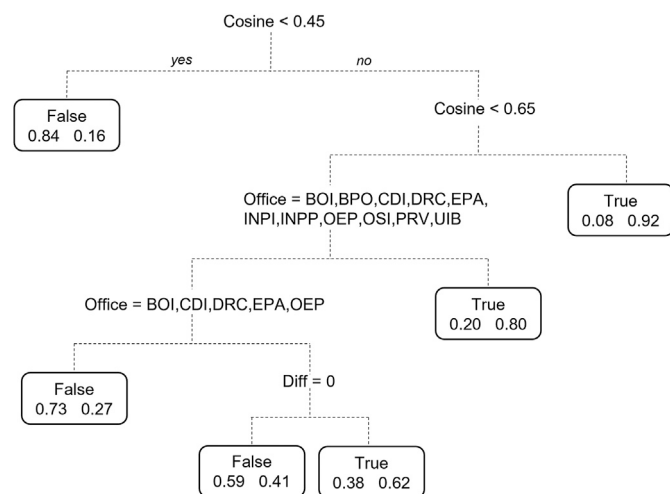


Fig. 16. Example of rules for the Decision Tree classifier with LLM3 embedding model fine-tuned with 10 epochs.

A second usage context is decision support for offices and courts. For examination and opposition at EUIPO and national offices, similarity matrices generated by the model can serve as an internal map of the case, while the underlying scores calibrated by office reveal divergences in practice (for example Malta versus EUIPO). Boards of appeal and courts may also use the models or approach describe here in a purely diagnostic way, to check internal consistency and to retrieve comparable earlier decisions.

6.2. Thresholds and evidence surfacing

The models output cosine similarities or classifier scores that approximate the probability that a pair of items is treated as similar by a given office. Different thresholds are appropriate depending on the goal. For example, users can select a high threshold on the score or cosine such that the precision is high on held-out data for the relevant office and quickly find a suitable pairs. Other practitioners may lower this threshold value to favor recall at the cost of additional false positives. Pairs above are highlighted for review, while pairs below remain accessible but are not prioritized. This mirrors clinical triage designs, where high-recall operating points are combined with systematic escalation of uncertain or high-risk cases to human clinicians [22].

Our models can also be combined with caselaw retrieval where the system can operate without a strict threshold. For each pair of items, it retrieves the k nearest neighbors in embedding space, filtered by office, class or origin where needed. Attorneys or examiners can request “EUIPO-style” neighbors or neighbors for a particular national office and directly compare convergent and divergent practice. This follows recent work on LLM-assisted patent tools, where summaries and similarity scores are used to surface relevant patents, while experts retain control over evaluation [23].

6.3. Human-in-the-loop safeguards and responsibility

When deployed inside a trademark office or a law firm, the system should be embedded as a human-in-the-loop tool. The EU AI Act classifies AI systems used in the administration of justice as high-risk; trademark examination by public authorities falls within this field. In such settings, the model must therefore be framed explicitly as an assistant for similarity analysis and evidence surfacing. Responsibility for any legal decision remains with the examiner, judge or attorney and, institutionally, with the deploying authority, whereas the model provider is responsible for the construction of the train dataset, validation procedures and technical documentation.

In practice, for each opposition, the system can generate a color-coded similarity matrix for all goods and services pairs, office-specific scores, and a ranked list of earlier decisions. Experienced users first review the pairs with the highest scores, inspect the associated item descriptions and precedents, and then either an examiner records an explicit decision with reasons or an attorney formulates written arguments citing the retrieved decisions. All interactions are logged, including cases where the human decision deviates from the system’s suggestion, so that model behavior can be audited and recalibrated over time. Disagreement between EUIPO-calibrated and national-office predictions, or scores in a predefined intermediate band, can be configured to trigger an additional review step by a second examiner.

These safeguards are consistent with human-in-the-loop architectures proposed for AI-assisted patent drafting and judicial AI assistants, where AI systems act as tools for drafting or search and legal professionals remain responsible for the content and outcome [24]; [25]; [26]. In future work, replacing dense embeddings with interpretable node representations would further align the system with legal reasoning requirements and regulatory expectations.

7. Conclusion

7.1. Theoretical and practical implications

In this work, we have proposed a range of solutions with the aim of improving the state-of-the-art in automatic similarity detection in trademarks.

The first of these is the use of a large language model fine-tuned with our data for the vectorization of items. We hereby submit a proposal for a formula that would facilitate the automatic selection of the threshold in order to reach an optimal balance between the two classes of classification errors. Choosing an LLM considerably increased the performance of the Doc2Vec or Word2Vec models used previously by [15] on the Nice 45 class as well as on other classes. We have also shown that by specializing models on our data, it is not necessary to use very large models; a small model such as LLM3 delivers equivalent performance.

The second model is the definition of a hybrid model integrating item embeddings, contextual features and interactions. The SHAP values confirm that integrating contextual variables, in particular office, considerably increases performance. The model is trained on pooled decisions from multiple offices so that it learns a shared representation of semantic relatedness, while contextual variables allow office-specific calibration of decisions. This makes institutional practice a first-class modeling object rather than an uncontrolled nuisance: the semantic core provides stability, and contextual variables provide transparent, auditable calibration that reflects real decision environments.

The use of a hybrid model, in contrast to other approaches based solely on embeddings [14], could also make it possible in the future to add images such as brand logos as model inputs, to combine our approach with the one of [1] or [7] as example.

And finally, we have proposed a third solution that enhances the baseline model while being lighter and more explainable than the hybrid MLP classifier.

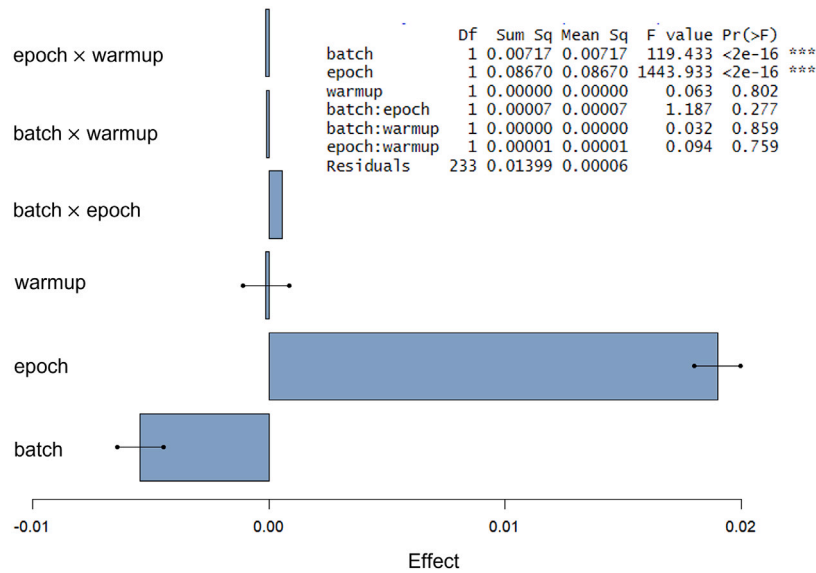
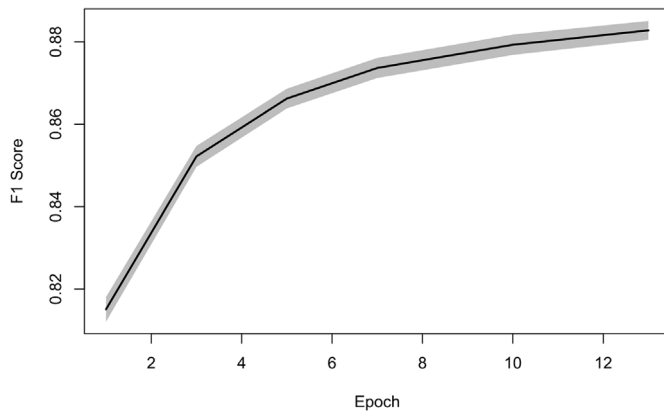
7.2. Future works

Explainability of the model is an essential point in the context of trademark opposition. Indeed, the rejection of a trademark application by a court or an office must be argued. When using SHAP values with the hybrid MLP model or a reduced explicable model, such as a decision tree, the features that explain the classification of an example will certainly be the office and embeddings. However, embeddings (Doc2Vec, Word2Vec or LLM) are vectors that cannot be interpreted. In the context of trademark application procedures, it is not possible to argue that rejection (or acceptance) is due to specific embeddings without understanding them. It would therefore be beneficial to replace large language models with an explicable vectorization model, such as that proposed by [27].

Table A.4

Mean and standard deviation of the f1 score computed with the new index given by (2) and the Youden's J index.

Model	New index	Youden's J index	p-value 1	p-value 2	p-value 3
d2v	0.434 ($7.8 * 10^{-3}$)	0.010 ($5.5 * 10^{-3}$)	$< 2.2e10^{-16}$	0.42	$< 2.2e10^{-16}$
w2v	0.667 ($6.1 * 10^{-3}$)	0.583 ($3.4 * 10^{-2}$)	$1.4 * 10^{-13}$	0.48	$3.1 * 10^{-9}$
LLM3	0.879 ($6.1 * 10^{-3}$)	0.879 ($6.7 * 10^{-3}$)	0.97	0.59	0.13
LLM12	0.888 ($7.3 * 10^{-3}$)	0.888 ($7.4 * 10^{-3}$)	0.92	0.13	0.11
Distil Roberta	0.885 ($7.1 * 10^{-3}$)	0.886 ($6.2 * 10^{-3}$)	0.85	0.81	0.16
mpnet	0.890 ($7.5 * 10^{-3}$)	0.889 ($7.6 * 10^{-3}$)	0.86	0.16	0.11
Stella	0.891 ($5.9 * 10^{-3}$)	0.893 ($2.4 * 10^{-3}$)	0.81	0.66	0.57

**Fig. B.17.** Hyperparameters effects with 95% confidence interval and ANOVA results.**Fig. B.18.** F1 score vs number of epochs for LLM3 with batch size=16 and warmup=0.01. The black line corresponds to the average value of the 30 cross-validation repetitions and the gray areas to the confidence interval.

CRedit authorship contribution statement

Yannick Le Nir: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Astrid Jourdan:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Cherif Sidhoum:**

Writing – review & editing, Resources. **Nicolas Girardin:** Writing – review & editing, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Index for the threshold selection

A comparison was conducted to compare the index given by Eq. (2) with the Youden's J index ($recall + specificity - 1$) [28] commonly used to determine the threshold. Table A.4 give the mean and the standard deviation of the F1 scores computed with the two indices and the p-values of several statistical tests.

The first (p -value 1) is to decide whether the difference between the F1 scores is statistically significant (H_0 : F1 score with the new index = F1 score with the Youden's J index). The p-values are very high for the five LLMs, so using one index or the other does not significantly affect the F1 scores for the LLM models. In contrast, the p-values of d2v and w2v are very small which indicates a significant difference between the F1 scores. The new index gives better F1 scores. We can also note the very low F1 score for d2v with Youden's index. As highlighted in Section 3, this is due to the distribution of cosine values. The threshold value defined by Youden's index is very high (0.9 for the Youden's

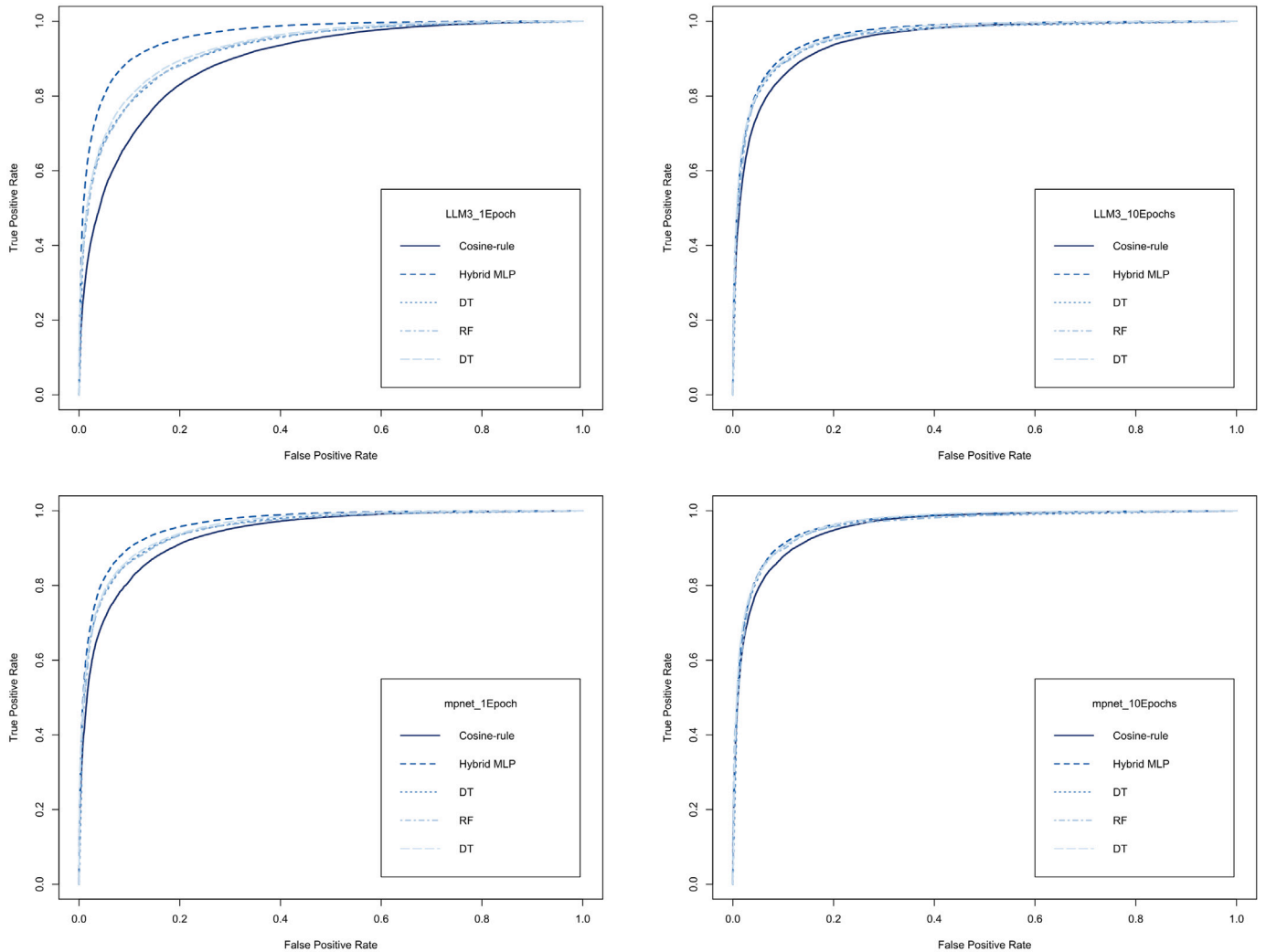


Fig. C.19. ROC curves of the F1 scores per embedding models for different classifiers.

index versus 0.5 for the new index), which significantly reduces recall in favor of specificity.

This conclusion is confirmed by the statistical test used to determine whether there is a balance between recall and specificity (H_0 : recall-specificity = 0) in the case of the new index (p -value 2) and Youden's index (p -value 3). For the five LLMs, all p -values are large enough to conclude that there is a balance between recall and specificity. However, it should be noted that the p -values of the new index are systematically higher than those of the Youden index, which means that the hypothesis of balance is more easily accepted with the new index. In the case of d2v and w2v, the p -values 2 are large enough to accept that the new index provides a balance between recall and specificity. However, the p -values 3 indicate a significant imbalance for the Youden index. As mentioned above, the Youden index selects very high thresholds for d2v, resulting in low recall and high specificity. The threshold selected for w2v is much lower, but it remains too high compared to the cosine values (0.37 for Youden's index versus 0.33 for the new index). So here again, recall is considered significantly lower than specificity. In conclusion, the new index provides better threshold choices than the Youden's J index. This conclusion applies only to our dataset, and a broader study will need to be conducted to confirm or refute it.

Appendix B. Selection of the hyperparameters

To determine the hyperparameters, we set up a complete factorial design for the three hyperparameters: batch size between 16 and 32, number of epochs between 1 and 10, and warmup between 0.1 and 0.01. The design includes 8 experiments integrated into the cross-validation process (Fig. 1), i.e. repeated 30 times. As expected, Fig. B.17 demonstrates that the F1 score increases when the batch size decreases, and more significantly when the number of epochs rises. Warmup has no impact on the F1 score alone or in interaction with other hyperparameters. Since the number of epochs has a significant impact, we studied the convergence curve beyond 10 epochs. Fig. B.18 shows that the curve continues to grow after 10 epochs, however a statistical test on the F1 score between 10 and 13 epochs shows that this increase is almost insignificant (p -value=0.05). For reasons of computation time, we therefore limited the number of epochs to 10.

Appendix C. Additional graphics

See Figs. C.19 and C.20.

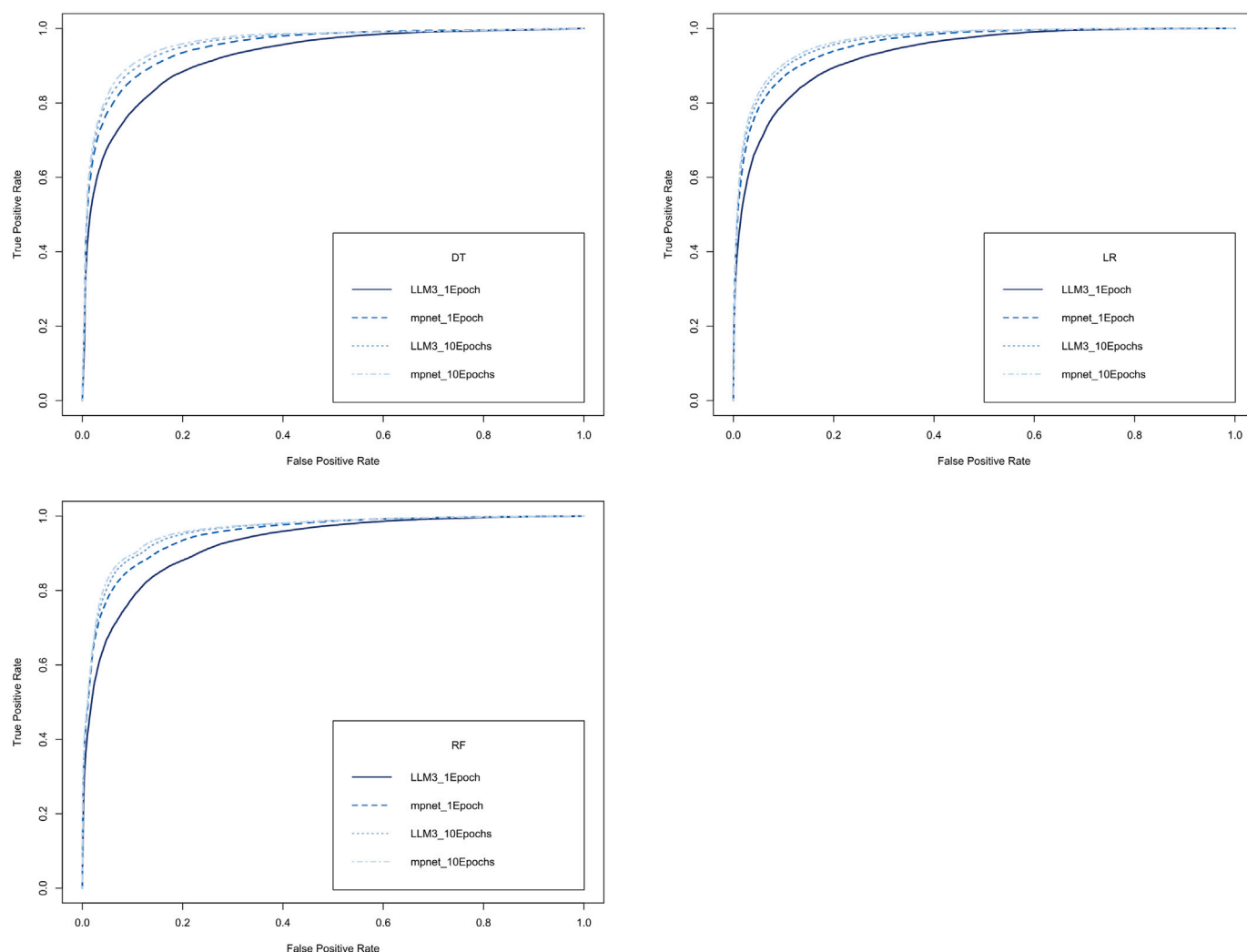


Fig. C.20. ROC curves of the F1 scores per classifier for different embedding models.

Data availability

Data will be made available on request.

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